Announcements

- Any questions/issues?
- Schedule:

  Today (Follow-up): Question 2 from "Climate Sensitivity" discussion — Bb curves and absorption lines
  Today (New): Timescales of Climate Change
  Tomorrow (Lab): Absorption by CO_2 (Part II).
  Thursday (Discussion): Timescales of Climate Change
  Next Week: Modeling Climate Change

Today's Reading Quiz (3 min)

On your own sheet of paper...

Briefly explain radiative forcing. What are its units and why? How does it affect temperature of the Earth? Can you give one example?

Turn in to Blackboard as scanned pdf (or image is fine).

Final Poster Project

- Choose a topic that is interesting to you and make a poster to present (5min) to the class during our final exam period (Thurs, Dec 11, 10am).
- Work on your own or with a partner.
- Think of it as a Discussion/Lab/Lecture that you create, explaining some topic to yourself and us.
- **Talk to me by Thanksgiving (Tues Nov 25)** to verify your project topic (**worth one Discussion assignment**).
- In the last week of class you will have time (lecture and discussion) to work on your final poster project.

Forcing: Adding an atmosphere instantaneously

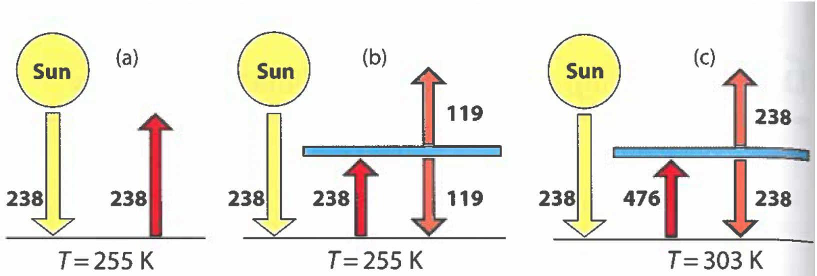


Figure 6.1 Schematic of energy fluxes on a planet (a) with no atmosphere, (b) the instant after a one-layer atmosphere is added to the planet, and (c) after the climate reaches its new equilibrium.

In this figure from Dessler, we see how an equilibrium Earth with no atmosphere (with an equilibrium temperature of 255 K due to $F_{\text{earth}} = F_{\text{sun}} \ \rightarrow \ \sigma T^4 = \frac{S}{4}(1-\alpha)$) abruptly changes to an out-of-equilibrium Earth with an atmosphere. The imbalance in the middle diagram exhibits a 119 W/m^2 **forcing** on the climate system. Equilibrium is restored in the final picture (our prior calculation of a single-atmosphere 30°C surface temperature).

Forcing: Seen in Blackbody curves

2. Now let's try to understand how CO_2 absorption lines lead to an increased overall temperature of the Earth.

   (a) Figure 2 shows simulated blackbody curves for the Earth's infrared emission, with different amounts of atmospheric CO_2. Describe generally how the absorption line changes as carbon dioxide is added to the atmosphere.

   (b) When does the CO_2 absorption line "saturate" and what does that mean?

   (c) On graph paper, draw an "approximate" blackbody curve as a (dashed-line) isosceles triangle with the following characteristics: base points at 5 μm and 25 μm; a peak of 23.8 $W/m^2/\mu m$. What is its area under the curve?

   (d) Make a carbon dioxide "absorption line" in your curve centered at 15 μm. Give it a depth that shows "saturation" (not all the way down to zero) and try to give it a width so that it "eats" $\frac{1}{3}$ to $\frac{1}{2}$ of the original area. What is the area of your triangle with the absorption line removed?

   (e) Try to draw another isosceles triangle (with same base) that also has an absorption line of the same width and (fractional depth), but has total area equal to the original (un-eaten) triangle.

Forcing: The effect on Temperature

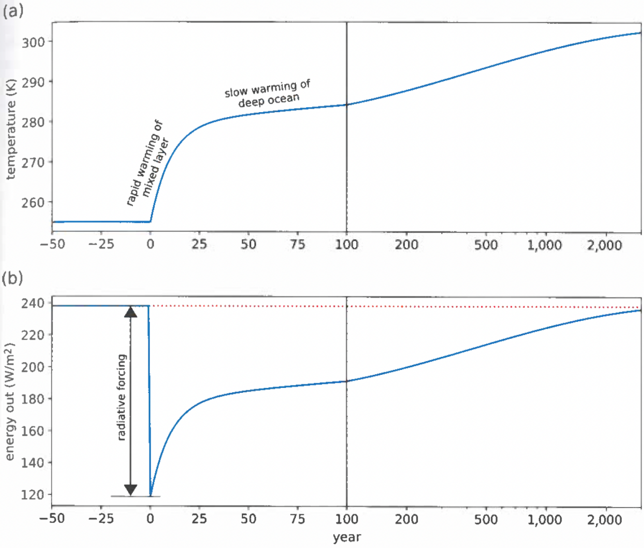


Figure 6.2 (a) Plot of temperature for the planet shown in Figure 6.1 as a function of time, (b) plot of *energy out* for the planet. A one-layer atmosphere is added instantaneously in year 0, and E_{in} for the planet remains constant at 238 W/m^2 (the dotted red line in panel b).

Definition of Radiative Forcing

Definition of **radiative forcing**:

and its effects on temperature:

Sources of Radiative Forcing

Net (anthropogenic) Radiative Forcing since the Industrial Revolution:

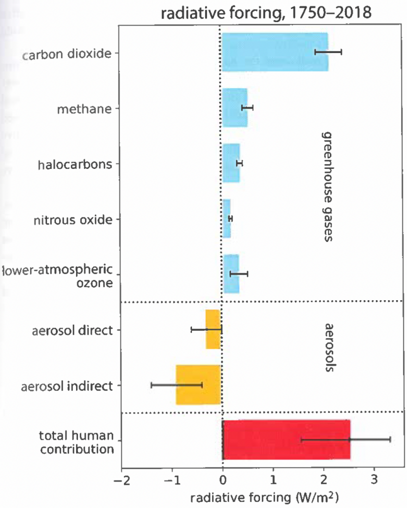
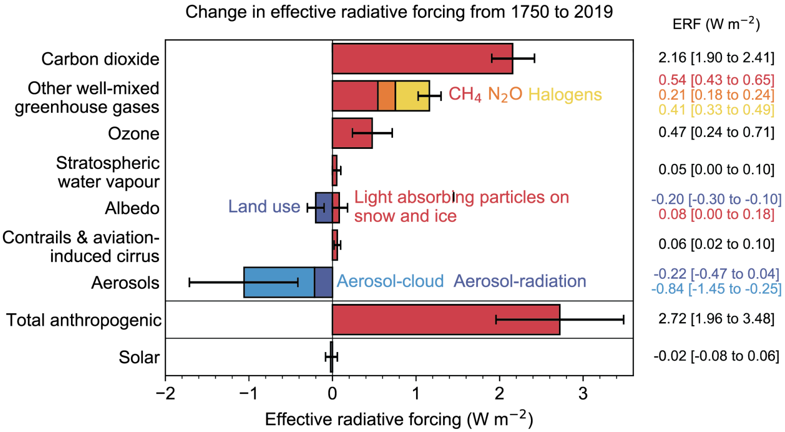


Figure 6.3 Radiative forcing caused by human activities between 1750 and 2018. The error bars indicate the uncertainty of the estimate.

Sources of Radiative Forcing

Net Forcing since the Industrial Revolution (IPCC):



The Effect of Aerosols

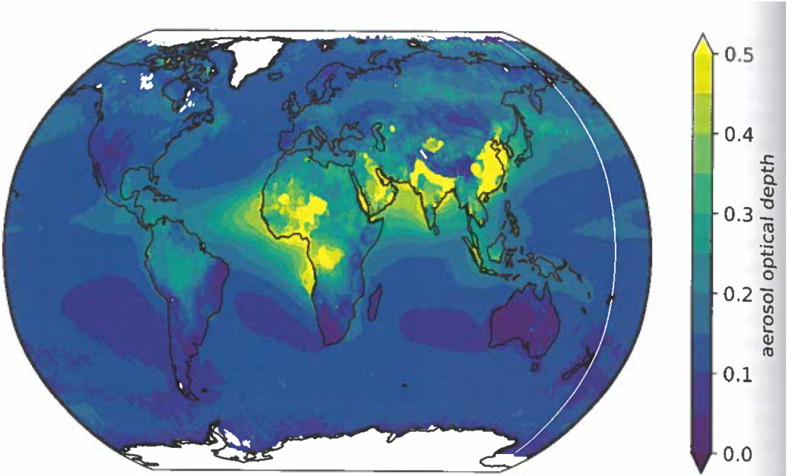


Figure 6.7 Global map of annual average aerosol optical depth (a measure of the abundance of aerosols) for the years 2005–2019. White areas denote regions where no data were obtained. Data are the MYD08 monthly aerosol fields from the Moderate Resolution Imaging Spectroradiometer onboard NASA's Aqua satellite. This plot was produced on May 25, 2020, using the Giovanni online data system (https://giovanni.gsfc.nasa.gov/giovanni/), developed and maintained by the NASA GES DISC.

Important Questions about Forcing